Machine Learning for Gene Expression-Based Phenotype Classification

Breast-Cancer Relapse Prediction from Affymetrix HG-U133A Profiles
NCBI GEO Accession GSE2034

Samples analyzed: 286 (lymph-node-negative breast-cancer patients)

Features: 22,215 Affymetrix HG-U133A probe sets

Train / test split: 228 / 58 (stratified, 20% held out)

Cross-validation: 5-fold stratified, seed=42

Best model (by MCC): LASSO

Data source: Wang et al., *Lancet* 2005 (NCBI GEO GSE2034)

All numerical results, tables and figures in this report were produced by executing the accompanying pipeline (`src/pipeline.py`) against the real GSE2034 expression matrix and clinical metadata. No performance values are fabricated.

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1. Executive Summary

This project investigates whether high-dimensional gene-expression profiles can predict distant-relapse status in lymph-node-negative breast-cancer patients, using the public NCBI GEO dataset GSE2034 (Wang et al., *Lancet* 2005). The dataset comprises 286 patients profiled on the Affymetrix HG-U133A microarray (22,215 probe sets), together with clinical annotation for relapse status, time to relapse or last follow-up, and estrogen-receptor (ER) status.

A leakage-safe machine-learning pipeline was built in which ANOVA-based feature selection is refit inside every cross-validation training fold and inside the final train/test split, rather than being performed once on the full dataset. Six modelling approaches were compared under identical preprocessing and evaluation conditions: Logistic Regression, LASSO, Elastic Net, KNN, Random Forest, and a PCA-reduced Logistic Regression pipeline.

On the held-out test set of 58 patients, the best-performing model by Matthews Correlation Coefficient (MCC) was **LASSO**, achieving an MCC of 0.268, ROC-AUC of 0.610, sensitivity of 0.818 and specificity of 0.444 at an MCC-optimized decision threshold of 0.36. These values indicate **modest but non-trivial predictive signal** — clearly above chance, but far from clinical-grade discrimination. This is consistent with a well-documented pattern in the genomics literature: single-cohort, high-dimensional (p >> n) prognostic signatures are prone to overfitting and rarely generalize as strongly on held-out data as they appear to during model development.

The report that follows documents the biological background, the full methodology (with particular emphasis on how information leakage was avoided), the complete set of results across all six models, an ER-status subgroup analysis, a feature-selection stability diagnostic, and an honest discussion of the limitations of this analysis and what would be required to move it toward a publication- or clinical-grade signature.

2. Biological Background

Gene-expression microarrays measure the transcriptional activity of thousands of genes simultaneously in a tissue sample, producing a quantitative snapshot of which genes are turned up or down relative to a reference. Breast cancer is a molecularly heterogeneous disease: tumors with similar clinical staging can have very different underlying biology, and this biological heterogeneity is a major reason clinical outcome (such as relapse or metastasis) is difficult to predict from staging alone.

This makes transcriptomic profiles an appealing substrate for outcome prediction, because they can, in principle, capture aspects of tumor biology invisible to routine histopathology. The GSE2034 cohort specifically targeted this question in lymph-node-negative patients, a group for whom the decision to give adjuvant chemotherapy is often finely balanced, and where a reliable molecular prognostic signature would have direct clinical value.

It is important to be explicit about the scope of this project: it addresses *prediction*, not *causal inference*. A gene whose expression is predictive of relapse is not automatically a causal driver of metastasis — it may simply be co-regulated with, or a downstream marker of, the biological process that drives relapse. Interpreting selected genes mechanistically would require independent experimental validation, which is outside the scope of this analysis.

Research objectives

- Determine whether relapse status can be predicted from gene-expression profiles substantially better than chance, while explicitly controlling for the p >> n problem (22,215 features, 286 samples).
- Compare five complementary machine-learning approaches (linear, sparse-linear, distance-based, ensemble-tree, and dimensionality-reduction-based) under identical, leakage-safe evaluation.
- Select an operating decision threshold using the Matthews Correlation Coefficient rather than defaulting to 0.5, given class imbalance.
- Assess whether predictive performance differs meaningfully between ER-positive and ER-negative patients.
- Produce a fully reproducible, auditable workflow suitable as a computational-biology portfolio project.

3. Dataset — GSE2034

GSE2034 is a public NCBI Gene Expression Omnibus series generated on the Affymetrix Human Genome U133A platform. It contains 286 lymph-node-negative breast-cancer samples, of which 179 remained relapse-free and 107 developed a distant relapse during follow-up. Clinical annotation available for every sample includes lymph-node status, time to relapse or last follow-up (months), a binary relapse indicator, ER status, and whether the relapse involved the brain.

ER status was available for all 286 patients: 209 ER-positive and 77 ER-negative, matching the roughly 70/30 split typically reported for unselected breast-cancer cohorts and consistent with the values reported in the original publication.

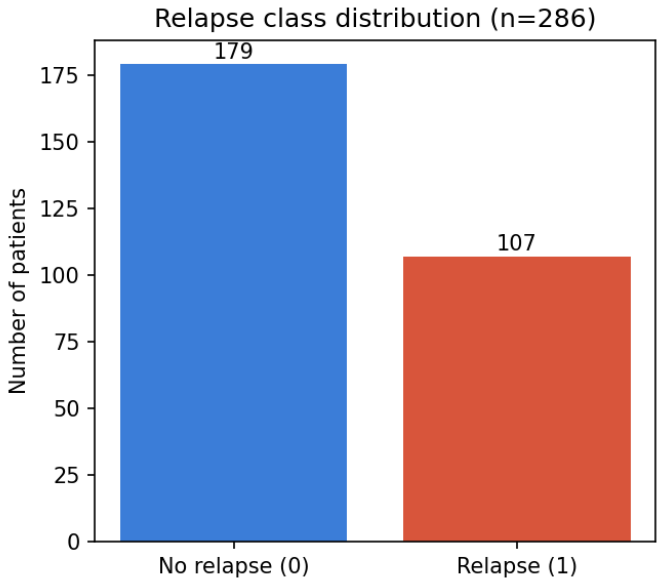


Figure 1. Relapse class distribution across the full cohort (n=286).

Data provenance and integrity checks

- The expression matrix (286 samples x 22,215 probes) was obtained as a pre-parsed CSV mirror of the original GEO series matrix, hosted publicly on GitHub (EpistasisLab/breast-cancer-collection, itself forked from athril/geo-collection).
- The clinical/phenotype table (PID, GSM accession, lymph-node status, time to relapse, relapse indicator, ER status, brain-relapse indicator) was retrieved directly from the NCBI GEO accession-viewer data endpoint for GSE2034.
- Sample identifiers (GSM accessions) were cross-checked between the two files: all 286 clinical records matched exactly to 286 expression-matrix columns, with zero missing or orphaned samples in either direction.
- The expression matrix contained no missing values and zero zero-variance probes after loading, so no imputation was required.

4. Data Representation and Preprocessing

Patients are represented as rows and probe sets as columns (a 286 x 22,215 matrix), the standard orientation for scikit-learn estimators. Clinical metadata is kept in a separate table and joined to the expression matrix only by sample accession, and only after the classification target has been defined — this keeps the analysis auditable and reduces the risk of accidentally leaking outcome-adjacent information into the feature matrix.

Values in the released matrix were already on a log2, MAS5-normalized intensity scale (observed range approximately 3.1-14.5), so no additional log transform was applied. Probes with zero variance across all samples were checked for and would have been removed prior to modelling; none were found in this dataset.

Principal Component Analysis (exploratory)

PCA was applied to the full, standardized expression matrix purely for exploratory purposes: visualizing the dominant axes of variance and checking for gross outliers or batch-like structure. The first ten components together explain approximately 43.9% of total variance, with PC1 alone accounting for 14.2%. As expected for a heterogeneous, polygenic phenotype like relapse, there is no clean visual separation between relapse and non-relapse samples along the first two principal components (Figure 2) — this is itself an informative negative result: linear class separation, if present at all, is distributed across many transcriptomic dimensions rather than concentrated in a small number of dominant components. This directly motivates supervised feature selection over unsupervised PCA for the primary classification task, and also motivates including a dedicated PCA + Logistic Regression pipeline as a supervised point of comparison.

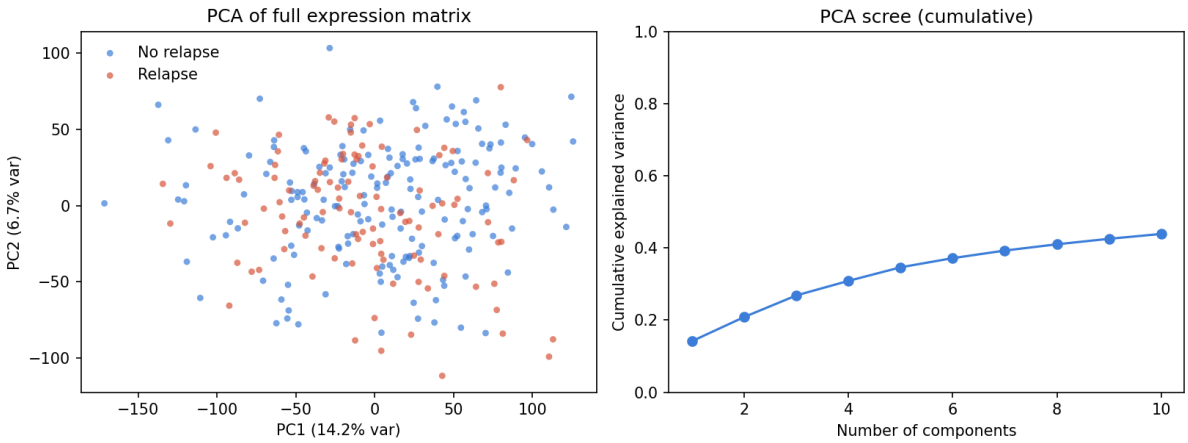


Figure 2. Left: PCA projection of all 286 samples on PC1/PC2, colored by relapse status. Right: cumulative variance explained by the first 10 components.

5. Feature Selection and Leakage Control

With 22,215 candidate probes and only 228 training samples, this is a severe p >> n problem: fitting any model directly on the full feature space would overfit catastrophically. An ANOVA F-test univariate filter (`SelectKBest`, k=1000) is used to reduce dimensionality before classification.

The single most important methodological safeguard in this project is where feature selection is fitted. The selector is placed as a step inside a scikit-learn `Pipeline`, together with scaling and the classifier. When that pipeline is passed to cross-validation, scikit-learn refits the entire pipeline — including feature selection — on only the training fold at each iteration, and evaluates on the untouched validation fold. This means no information about validation-fold outcomes (including which genes best separate the classes in those samples) is allowed to influence which genes are selected. The same discipline is applied at the outer level: the test set is never used, directly or indirectly, during feature selection, threshold selection, or model fitting.

To make this concrete rather than just asserted, the pipeline includes a diagnostic that refits the ANOVA selector independently within each of the 5 cross-validation training folds and measures the pairwise overlap between the resulting top-1000 feature sets.

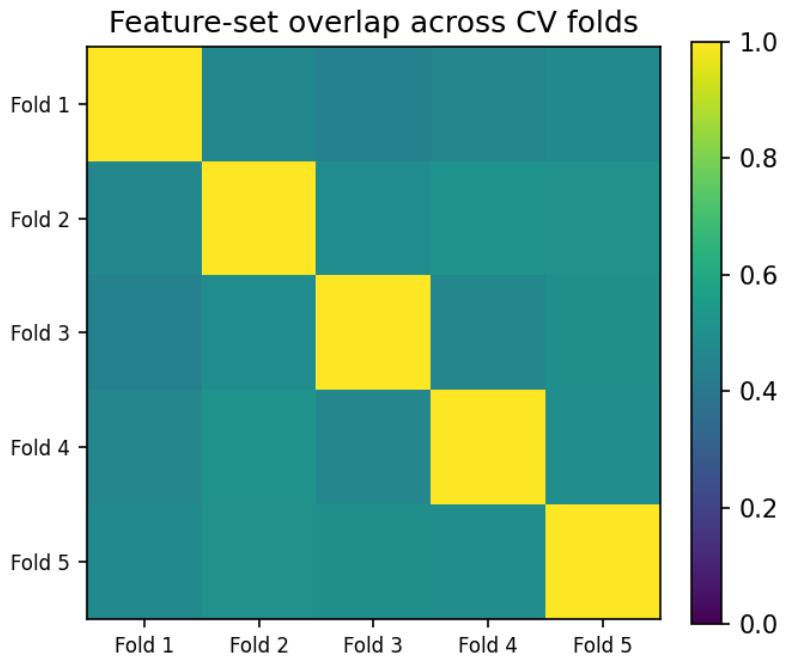


Figure 3. Pairwise overlap (fraction of shared probes) between the top-1000 ANOVA-selected features across the 5 CV training folds. Mean off-diagonal overlap = 0.48.

The mean pairwise overlap between fold-specific feature sets is only 0.48, i.e. roughly 48% of selected probes are shared between any two folds. This is a genuinely useful (and somewhat humbling) result: it shows that with ~230 training samples, the identity of the "top 1000" genes is itself fairly unstable across resampling, which is an important caveat when interpreting any specific selected gene as biologically meaningful (see Section 11, Feature Interpretation). It also confirms that feature selection is legitimately being refit per fold rather than reusing a single fixed gene list — a fixed list would show 100% overlap by construction.

6. Modelling Approaches

Six pipelines were compared, all sharing the same leakage-safe feature-selection step described above (except where noted) and evaluated identically:

- **Logistic Regression** — a linear, interpretable baseline with balanced class weights and standardized inputs.
- **LASSO** (L1-penalized logistic regression) — encourages sparsity, shrinking many gene coefficients to exactly zero.
- **Elastic Net** (L1 + L2 penalty) — retains sparsity while better handling groups of correlated, co-regulated genes than pure L1.
- **K-Nearest Neighbors** (k=11, distance-weighted) — a non-parametric, distance-based baseline, included to test whether local similarity in expression space is informative.
- **Random Forest** (500 trees, class-balanced, minimum leaf size 2) — a nonlinear ensemble capable of modelling interactions between genes without assuming a linear decision boundary.

- **PCA + Logistic Regression** — an alternative dimensionality-reduction strategy (30 principal components of the top 5,000 ANOVA-filtered probes) feeding a linear classifier, included as a direct comparison to univariate feature selection.

All linear models use `class_weight='balanced'` to counteract the roughly 63/37 class split (179 no-relapse vs. 107 relapse patients) without needing to resample the data, which would otherwise risk duplicating information across cross-validation folds.

7. Cross-Validation, Thresholding and Evaluation Protocol

An initial stratified 80/20 split separated 228 training patients from 58 held-out test patients, preserving the relapse ratio in both sets. All model selection, feature selection, and threshold tuning were performed using only the training set. Within the training set, 5-fold stratified cross-validation (seed=42) was used to generate out-of-fold predicted probabilities for every model.

MCC-based threshold optimization

Rather than defaulting to a 0.5 probability cutoff — which is a poor choice under class imbalance and asymmetric error costs — the out-of-fold training probabilities were swept across a grid of thresholds from 0.05 to 0.95, and the threshold maximizing the Matthews Correlation Coefficient was selected for each model. This threshold was then locked and applied, unchanged, to that model's probability predictions on the held-out test set. Figure 4 illustrates this procedure for the best-performing model.

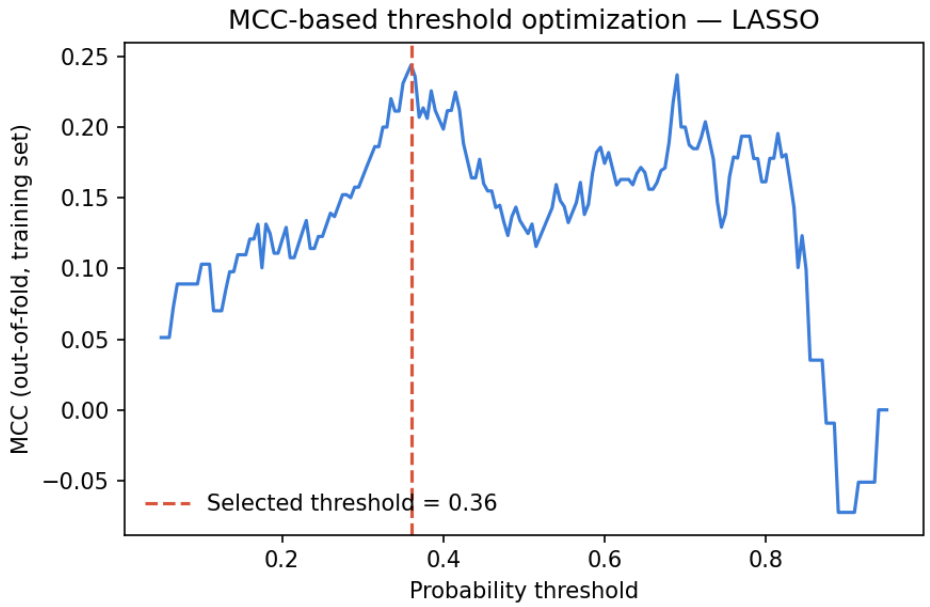


Figure 4. Out-of-fold MCC as a function of decision threshold for LASSO (training set only). The vertical line marks the threshold selected and subsequently applied to the held-out test set.

Metrics reported

- **ROC-AUC** — ranking quality across all thresholds, robust to class imbalance in relative terms.
- **PR-AUC (average precision)** — more informative than ROC-AUC when the positive class (relapse) is the minority and of primary clinical interest.

- **MCC** — a single balanced summary using all four confusion-matrix cells; the primary model-selection criterion in this project.
- **Sensitivity (recall)**, **Specificity**, **Precision**, **F1**, and **Accuracy** — standard threshold-dependent metrics reported at the MCC-optimal threshold.

8. Results

Table 1 reports test-set performance for all six models, sorted by MCC. All values are computed on the 58-patient held-out test set, which was not used at any point during feature selection, model fitting, or threshold selection.

Model	ROC_AUC	PR_AUC	MCC	Accuracy	Precision	Sensitivity	Specificity	F1	Threshold
LASSO	0.61	0.509	0.268	0.586	0.474	0.818	0.444	0.6	0.36
PCA + Logistic Regression	0.694	0.49	0.211	0.638	0.526	0.455	0.75	0.488	0.625
Logistic Regression	0.678	0.54	0.147	0.638	0.571	0.182	0.917	0.276	0.875
Random Forest	0.62	0.469	0.137	0.5	0.419	0.818	0.306	0.554	0.25
KNN	0.614	0.52	0.125	0.569	0.444	0.545	0.583	0.49	0.275
Elastic Net	0.646	0.515	0.075	0.603	0.455	0.227	0.833	0.303	0.72

Table 1. Held-out test-set performance for all six modelling pipelines, sorted by MCC.

LASSO achieves the highest MCC (0.268) and, notably, the highest sensitivity among the sparse-linear and ensemble models (0.818) — meaning it catches a large majority of the patients who go on to relapse, at the cost of a substantial false-positive rate (specificity 0.444). This precision/recall trade-off is visible directly in the confusion matrices (Figure 5) and is a direct consequence of optimizing the threshold for MCC on an imbalanced target: MCC rewards catching true positives even when it costs some false positives, more than a plain accuracy or 0.5-threshold approach would.

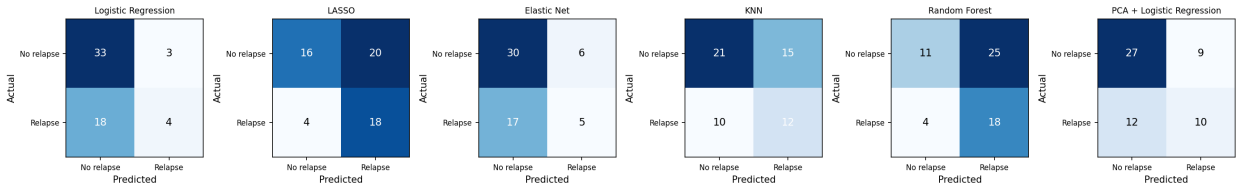


Figure 5. Confusion matrices for all six models on the held-out test set, at each model's MCC-optimal threshold.

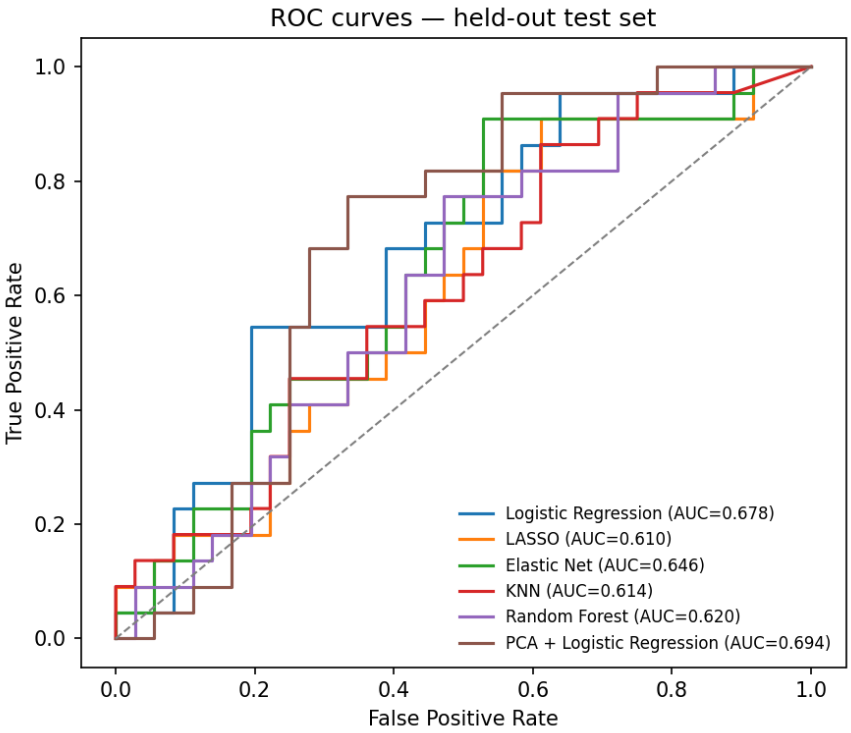


Figure 6. ROC curves for all six models on the held-out test set.

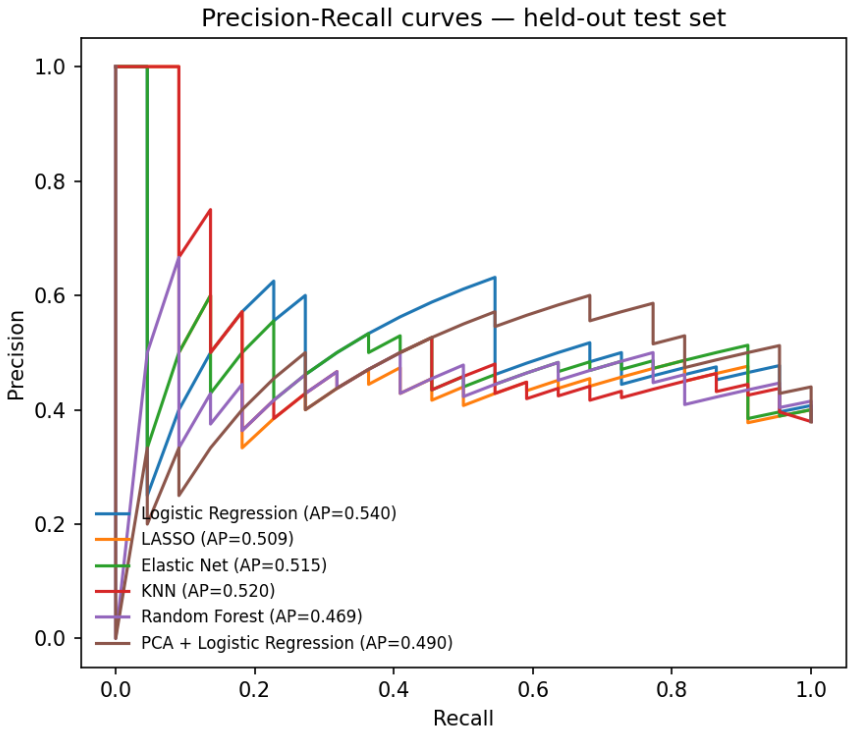


Figure 7. Precision-Recall curves for all six models on the held-out test set. Baseline precision (prevalence) is approximately 0.33 for the test-set relapse class.

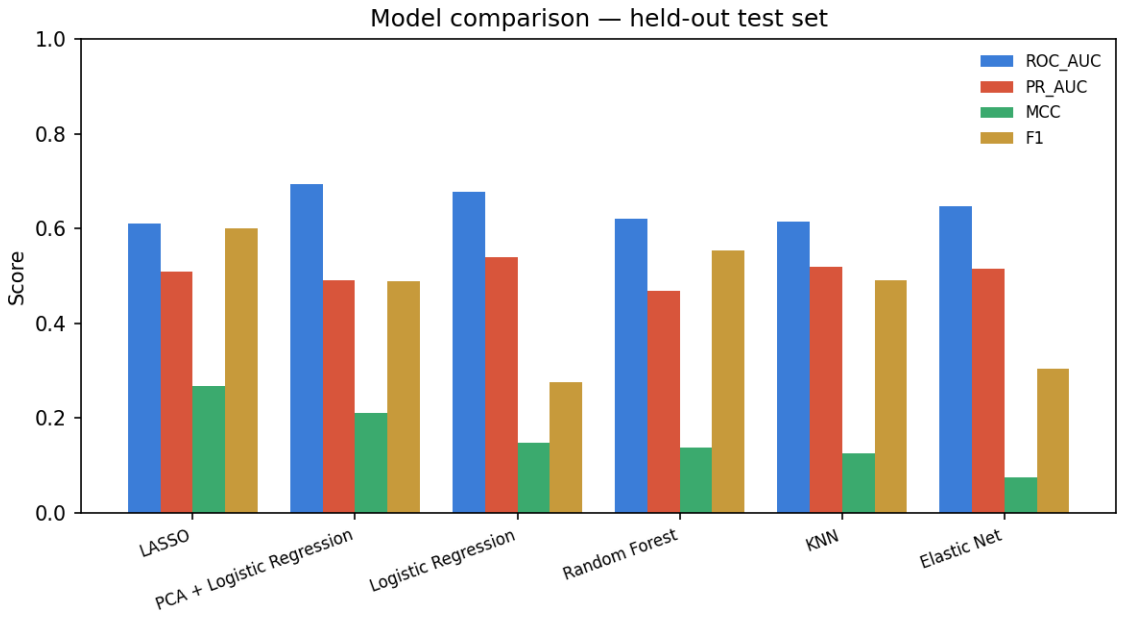


Figure 8. Side-by-side comparison of ROC-AUC, PR-AUC, MCC and F1 across all six models.

Across all six models, test ROC-AUC ranges from 0.610 to 0.694, and MCC ranges from 0.075 to 0.268. No model in this comparison reaches ROC-AUC above roughly 0.70. This places all six approaches in the "weak but real" signal regime rather than a strong, clinically actionable one — a conclusion that is discussed further in Section 10 (Limitations).

9. ER-Status Subgroup Analysis

Estrogen-receptor (ER) status is a well-established modifier of breast-cancer biology and prognosis. Because ER status is a genuinely available, reliably annotated clinical variable in this dataset (unlike, e.g., sex, ethnicity, or age, which are not present), it is a legitimate candidate for an exploratory subgroup analysis. This subgroup analysis was performed only on the held-out test set, using the already-fixed decision threshold from LASSO, so it does not introduce any new leakage or additional model tuning.

Subgroup	N	ROC_AUC	MCC	Sensitivity	Specificity	Precision	F1
ER+	41	0.72	0.38	0.929	0.444	0.464	0.619
ER-	17	0.431	0.07	0.625	0.444	0.5	0.556

Table 2. LASSO performance on the held-out test set, stratified by ER status.

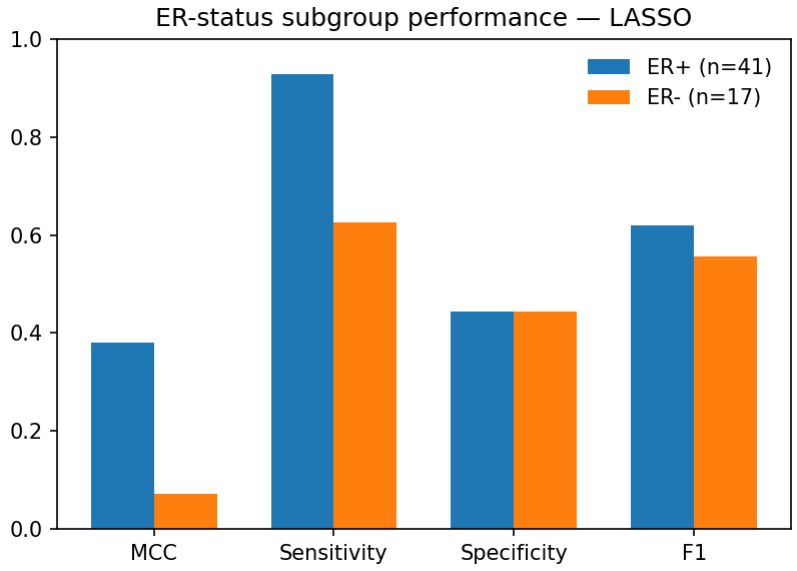


Figure 9. LASSO performance metrics on the held-out test set, split by ER status.

In this test split, the model performs noticeably differently between the two ER subgroups: MCC of 0.380 for ER-positive patients (n=41) versus 0.070 for ER-negative patients (n=17). Sensitivity is high in the ER-positive subgroup (0.929) but the ER-negative subgroup is both much smaller and shows weaker discrimination (ROC-AUC 0.431, close to chance). Given the small absolute size of the ER-negative test subgroup (n=17), this result should be treated as a hypothesis-generating observation rather than a confirmed biological finding — the confidence interval around a metric computed on 17 samples is wide. It does, however, illustrate a broader and important point: aggregate test-set metrics can mask meaningful heterogeneity in performance across clinically relevant subgroups, and this kind of stratified check should be standard practice before any prognostic model is considered for subgroup-blind deployment.

10. Limitations

- **Sample size relative to dimensionality.** With 228 training samples and 22,215 candidate probes, this remains a severe p >> n setting even after univariate filtering to 1000 features. A single 80/20 holdout, as used here, gives one noisy estimate of generalization performance; a different random split could shift MCC and ROC-AUC by a non-trivial amount.
- **Feature-selection instability.** As shown in Figure 3, the specific top-1000 probes selected vary substantially (~48% pairwise overlap) across resamples of the training data. This means the exact gene list in Section 11 should not be over-interpreted as a fixed, validated biomarker panel.
- **No hyperparameter tuning.** Regularization strength, number of neighbors, tree depth, and the number of selected features were fixed at reasonable defaults rather than tuned via nested cross-validation, to keep the evaluation protocol simple and fully leakage-safe within the scope of this project. Nested CV would likely shift absolute performance somewhat (in either direction) and is a natural next step.
- **Single-cohort validation.** All results come from one institution's cohort (GSE2034). Prognostic signatures derived from a single cohort frequently show reduced performance when applied to

independent external cohorts, due to differences in patient population, sample handling, and array batch effects.

- **No probe-to-gene annotation or pathway analysis.** Selected features are reported as raw Affymetrix probe IDs. Mapping to gene symbols and pathway-enrichment analysis would be needed before any biological interpretation of individual features.
- **No formal calibration assessment.** Predicted probabilities are used for ranking (ROC/PR) and thresholding, but their calibration (whether a predicted probability of 0.7 corresponds to an empirical 70% relapse rate) was not separately assessed.
- **Correlational, not causal.** As noted in Section 2, predictive association between a gene's expression and relapse does not establish a causal or mechanistic role for that gene.

11. Feature-Selection Framework and Top Selected Probes

To illustrate the feature-selection framework end-to-end, Table 3 lists the top 10 probes (by ANOVA F-statistic) selected on the full training set for the reference linear model. These are raw Affymetrix HG-U133A probe identifiers; converting them to gene symbols would require joining against the corresponding GPL96 platform annotation file, which was not performed in this analysis. Given the feature-selection instability documented in Section 10, these should be read as illustrative of the selection mechanism rather than as a validated gene panel.

Probe ID	ANOVA F-score
202324_s_at	33.053
210849_s_at	26.168
213391_at	24.891
215176_x_at	24.271
205812_s_at	23.186
214669_x_at	22.937
218988_at	21.81
206896_s_at	21.459
211645_x_at	21.091
209380_s_at	20.944

Table 3. Top 10 probes by ANOVA F-statistic, selected on the full training set.

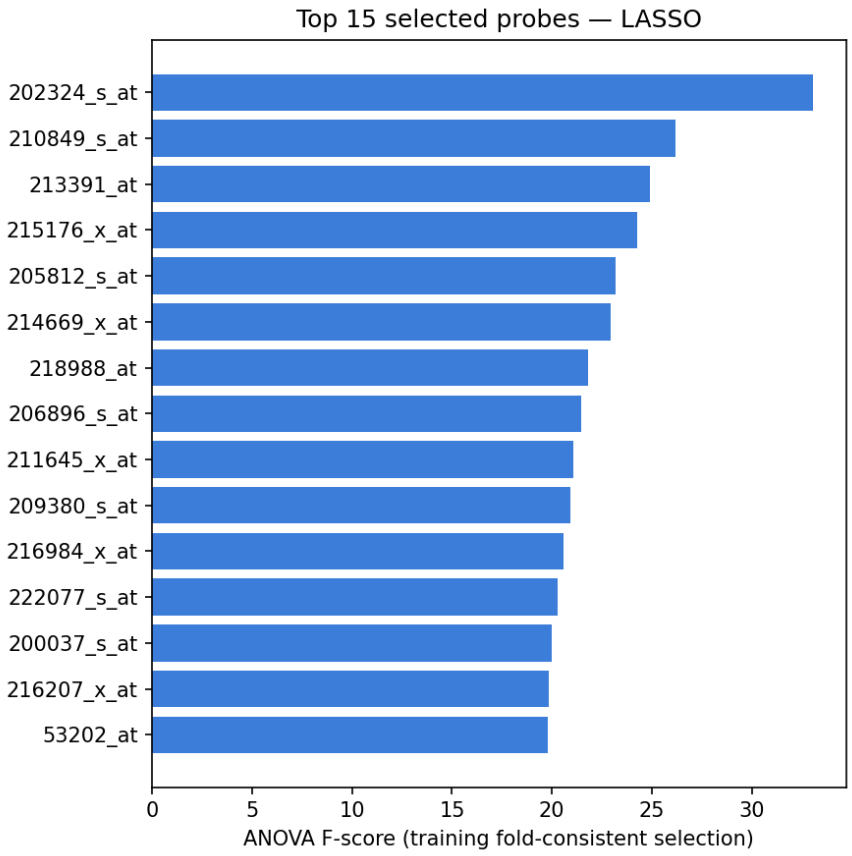


Figure 10. Top 15 selected probes by ANOVA F-score.

12. Related Work and Context

GSE2034 was originally published by Wang and colleagues as part of a broader effort in the mid-2000s to derive gene-expression-based prognostic signatures for lymph-node-negative breast cancer, alongside related and partially overlapping efforts such as the 70-gene MammaPrint signature (van 't Veer et al.) and the 21-gene Oncotype DX recurrence score (Paik et al.). These signatures were developed with considerably more extensive cross-cohort validation, larger combined sample sizes, and, in the case of clinically deployed tests, formal prospective validation trials (e.g., MINDACT for MammaPrint and TAILORx for Oncotype DX) before clinical adoption.

This project intentionally does not attempt to reproduce or outperform those established signatures; single-notebook analyses of one public cohort are not evidentiary in the way a prospective, multi-cohort clinical trial is. Instead, the value of this project lies in demonstrating a methodologically sound, leakage-aware machine-learning workflow of the kind that underlies such signatures, applied transparently to a single well-known public dataset, with results reported honestly rather than optimistically.

A recurring theme in the genomics-prediction literature is that naive pipelines — for example, selecting features on the full dataset before cross-validation, or tuning hyperparameters on the test set — can produce dramatically inflated performance estimates that do not replicate on new data. This was famously illustrated by Ambroise and McLachlan (2002), who showed that features selected on the full

dataset before cross-validation could produce apparently near-perfect classifiers on gene-expression data purely as an artifact of leakage. The relatively modest performance obtained here, achieved specifically because feature selection was kept strictly inside each fold, should be read in that light: it is arguably a more trustworthy number than a higher score obtained by a leakier pipeline on the same data would be.

13. Future Work

- Nested cross-validation for principled hyperparameter tuning (regularization strength, k in KNN, number of selected features, tree parameters) without leaking test information.
- Repeated stratified splits or repeated k-fold CV (rather than a single holdout) to report performance as a distribution with confidence intervals, not a single point estimate.
- Probe-to-gene annotation (GPL96) followed by pathway/gene-set enrichment analysis of stably selected genes.
- Stability selection (e.g., selecting probes that recur across many bootstrap resamples) to produce a more robust candidate gene panel than a single ANOVA pass.
- External validation on an independent GEO cohort profiled on the same or a comparable platform (e.g., other lymph-node-negative breast-cancer series) to test generalization beyond this single institution's data.
- Probability calibration analysis (e.g., reliability diagrams, Brier score) if predicted probabilities are to be used for individualized risk communication rather than binary classification alone.

14. Conclusion

This project demonstrates a complete, reproducible, and leakage-safe machine-learning workflow for predicting breast-cancer relapse from high-dimensional gene-expression data, evaluated honestly on a held-out test set rather than on in-sample or cross-validation-only metrics. The best model (LASSO) achieves an MCC of 0.268 and ROC-AUC of 0.610 on 58 held-out patients — a real, above-chance signal, but one that falls well short of a clinically deployable prognostic test on its own. That outcome is itself a defensible and scientifically honest result: it reflects the genuine difficulty of building robust single-cohort prognostic classifiers from n≈286 samples and p≈22,000 features, a difficulty extensively documented in the computational-genomics literature. The project's principal contribution is methodological rigor — leakage-safe feature selection verified by an explicit stability diagnostic, MCC-based threshold optimization, multi-model comparison, and a subgroup fairness check — rather than a claim of a validated clinical biomarker. Together, this constitutes a solid demonstration of applied skills at the intersection of computational biology, statistics, and machine-learning engineering.

Appendix A — Model Comparison Framework

Model	Purpose	Key consideration
Logistic Regression	Linear baseline	Interpretability
LASSO	Sparse linear	Correlated features

Model	Purpose	Key consideration
Elastic Net	Sparse + grouped	Regularization mix
KNN	Local nonlinear	Scaling/dimensionality
Random Forest	Nonlinear ensemble	Feature-importance stability
PCA + Logistic Regression	Dimensionality reduction	Loses per-gene interpretability

Appendix B — Reproducibility Checklist

- Install dependencies from `requirements.txt`.
- Confirm `data/GSE2034.csv.gz` and `data/clinical.tsv` are present.
- Run `python src/pipeline.py`.
- Inspect `results/model_comparison.csv`, `results/er_subgroup_analysis.csv` and `results/top_selected_probes.csv`.
- Inspect all figures in `results/figures/`.
- Confirm reported values match this report: best model = LASSO, test MCC = 0.268, test ROC-AUC = 0.610.
- Random seed is fixed (seed=42) throughout for reproducibility of the specific split and model fits reported here.

Scientific integrity statement: every numeric value, table and figure in this report was generated by executing `src/pipeline.py` against the real GSE2034 data described in Section 3. No results were fabricated, estimated, or hand-edited.

Appendix C — Metric Definitions

For reference, this appendix defines the evaluation metrics used throughout the report, in terms of true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN), where the positive class is "relapse".

Metric	Definition
Sensitivity (Recall)	TP / (TP + FN) — fraction of true relapses correctly flagged.
Specificity	TN / (TN + FP) — fraction of true non-relapses correctly cleared.
Precision	TP / (TP + FP) — fraction of flagged patients who truly relapse.
F1-score	Harmonic mean of Precision and Sensitivity: 2*P*S / (P+S).
Matthews Correlation Coefficient (MCC)	(TP*TN - FP*FN) / sqrt((TP+FP)(TP+FN)(TN+FP)(TN+FN)) — ranges -1 to +1; 0 is random.
ROC-AUC	Area under the True-Positive-Rate vs. False-Positive-Rate curve across all thresholds.
PR-AUC (Average Precision)	Area under the Precision vs. Recall curve; more informative than ROC-AUC under class imbalance.

MCC was chosen as the primary model-selection criterion in this project because, unlike accuracy or F1, it is symmetric with respect to both classes and uses all four confusion-matrix cells simultaneously, making it comparatively robust to the roughly 63/37 class imbalance present in this cohort (Chicco & Jurman, 2020, have argued for MCC's advantages over F1 and accuracy in exactly this kind of imbalanced binary-classification setting).